

Web & State Management

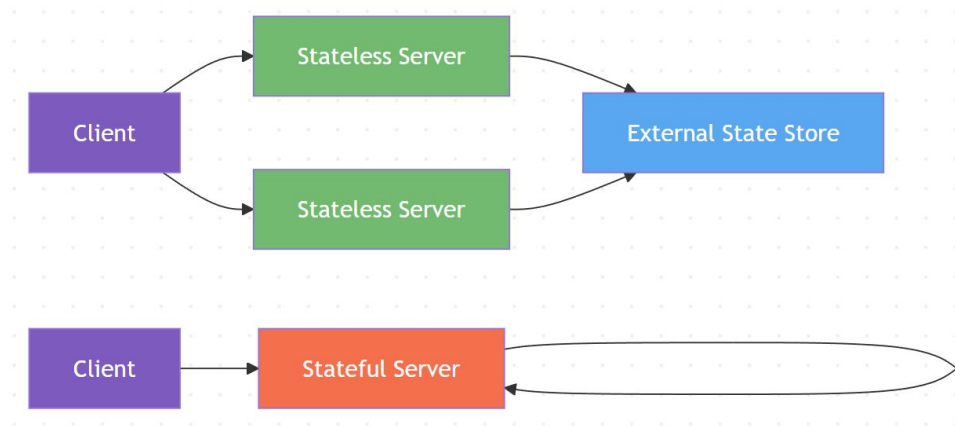
System Design Crash Course: Quick Revision Before Interviews

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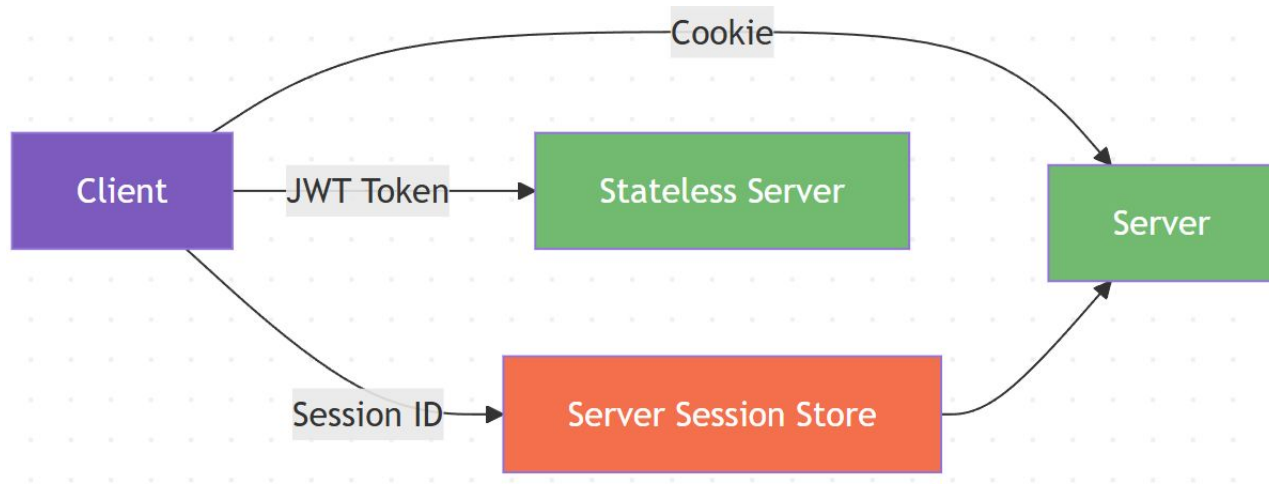
Stateless vs Stateful Systems

- Stateless: no server-side memory
- Stateful: server holds client state
- Stateless scales horizontally easily
- Stateful needs affinity or coordination
- State location drives architecture



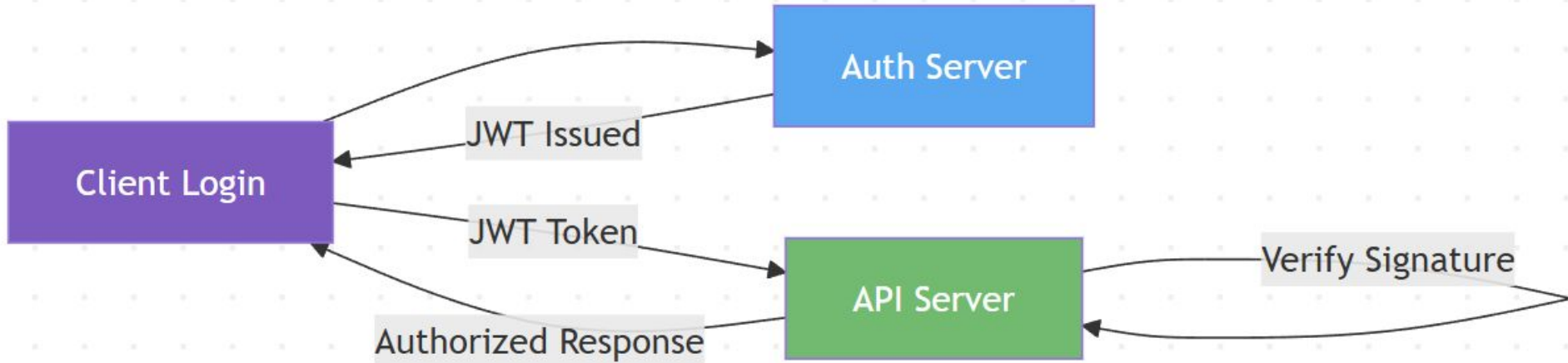
Cookies vs Sessions vs JWT

- Cookies store data on client
- Sessions store state on server
- JWT stores signed state client-side
- Sessions limit horizontal scaling
- JWT trades revocation for scale



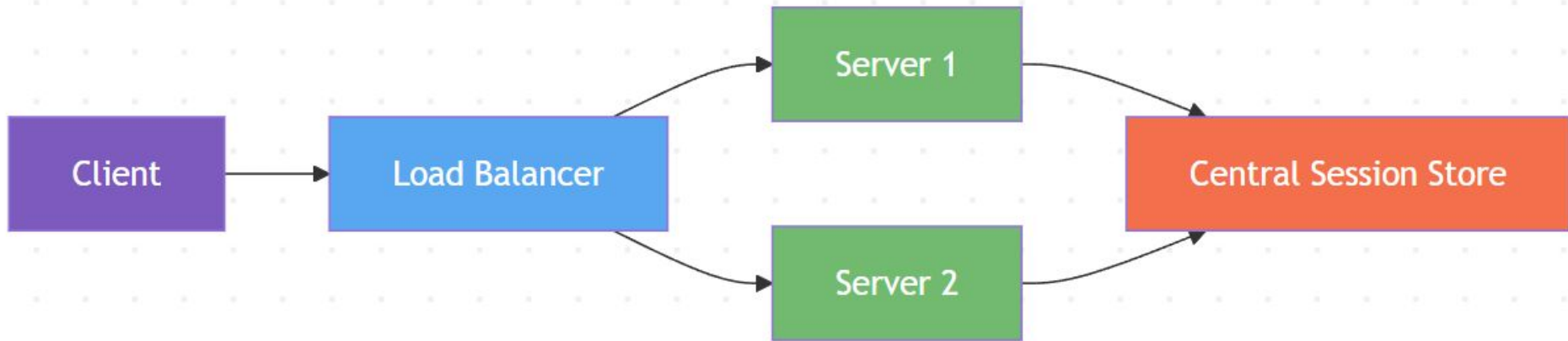
JWT and Stateless Authentication

- Token carries user identity
- Signed, not stored server-side
- Sent with every request
- Enables stateless servers
- Hard to revoke tokens



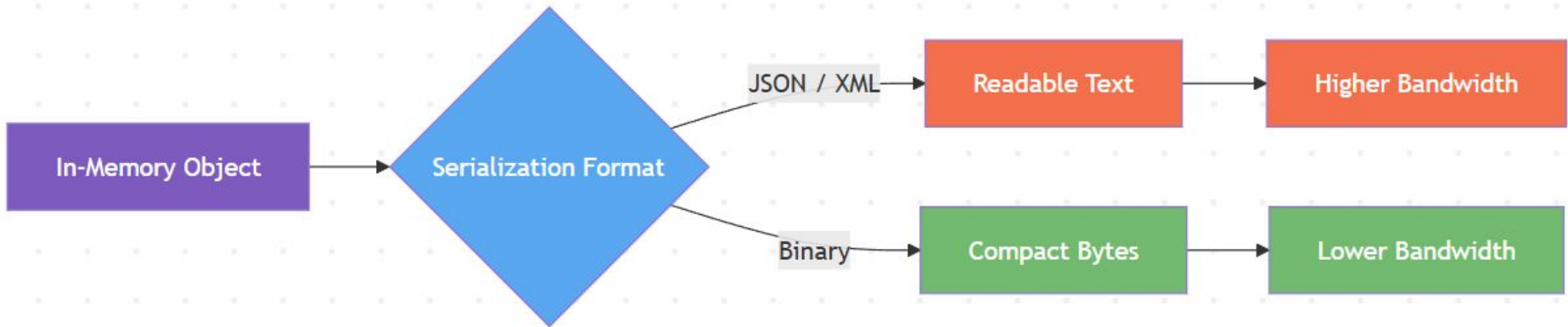
Session Storage in Distributed Systems

- Sessions store server-side user state
- Centralized session store required
- Avoid sticky load balancing
- Enables horizontal scaling
- Adds latency and dependency



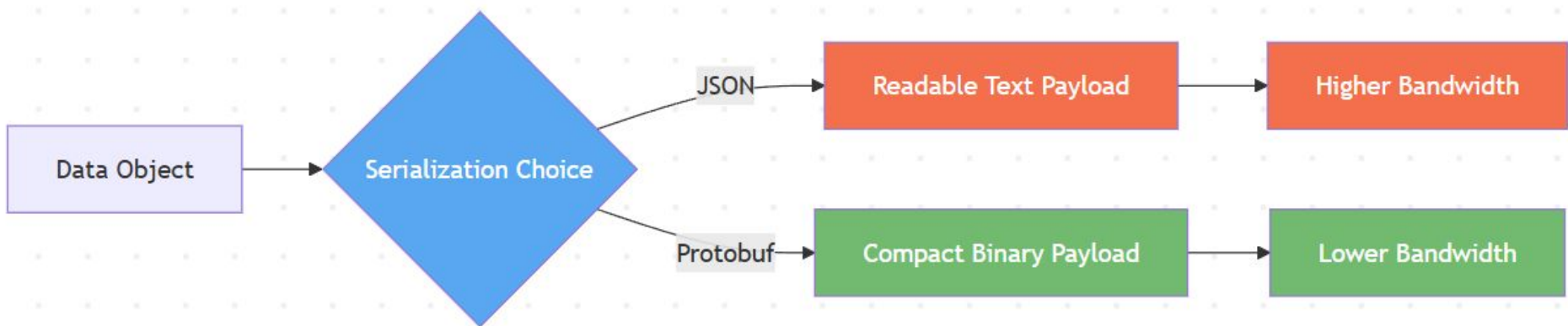
Serialization Basics

- Converts objects to transferable format
- JSON, XML are text-based
- Binary formats are compact
- Trade readability for bandwidth
- Impacts latency and performance



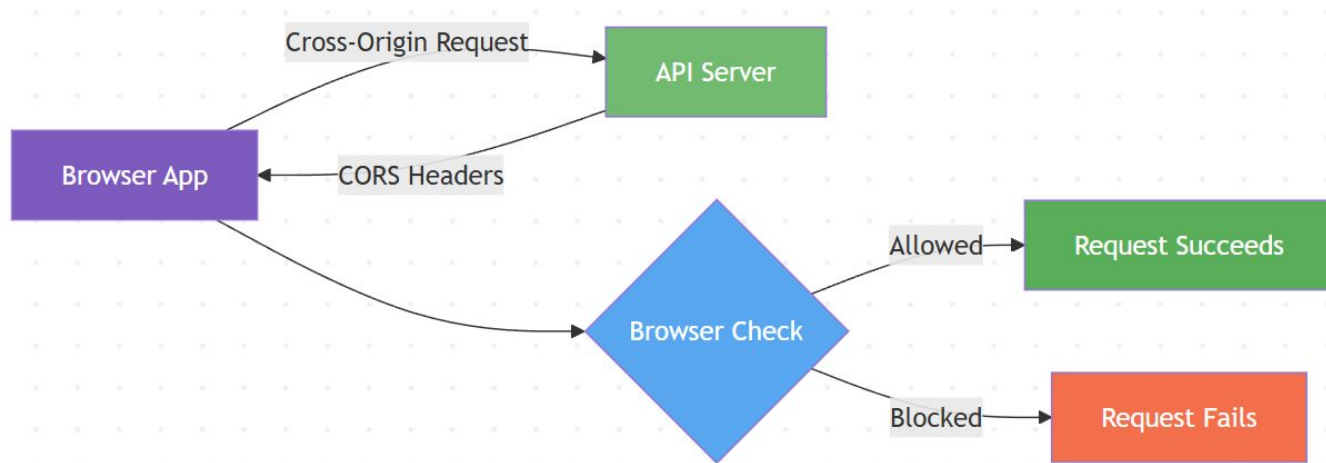
JSON vs Protocol Buffers

- JSON is human-readable text
- Protobuf is compact binary
- Protobuf smaller payloads
- JSON easier debugging
- Internal vs external API choice



CORS Explained

- Browser enforces same-origin policy
- Cross-origin requests blocked by default
- Server sends CORS headers
- Browser decides allow or block
- Security vs flexibility trade-off



Common CORS Pitfalls

- Confusing CORS with server security
- Missing preflight OPTIONS handling
- Using wildcard with credentials
- Forgetting allowed headers
- Debugging server instead browser

